

# SEGMENTATION OF AGRICULTURAL IMAGES USING CNN

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## ABSTRACT

This project investigated semantic segmentation of agricultural field imagery using a limited dataset of 50 labelled image pairs. A baseline convolutional neural network (CNN) was implemented using shallow convolutional architecture to classify pixels into background, crop, and weed classes. Images were resized during training to improve computational efficiency, while segmentation outputs were evaluated on a validation set. Baseline performance achieved a mean Intersection over Union (IoU) of 0.573 and global accuracy of 96.3%. Results showed strong background classification but reduced performance on crop and weed classes due to class imbalance and limited model depth. Further improvements with increased network depth and data augmentation techniques are then explored with an improved model, which saw an overall increase across all metrics and higher class-specific accuracy, including a global accuracy of 97.0%.

## 1. INTRODUCTION

This coursework investigates semantic segmentation under a constrained setting using only 50 labelled image pairs. The objective is to design and evaluate convolutional neural network (CNN) models capable of segmenting images into three classes: background, crop, and weed.

A baseline model is first developed using a simple fully convolutional architecture, to establish a reference level of performance and act as a control. This is followed by an improved model that increases network capacity and uses data augmentation to better utilise the limited dataset. The performance of both models is evaluated using standard segmentation metrics, including Intersection over Union (IoU), accuracy, and boundary-based measures.

The results highlight the limitations of naive architectures and demonstrate that improved model design and training strategies can significantly enhance segmentation performance, even in scenarios where data is limited.

## 2. METHOD

### 2.1. Dataset preparation

The supplied dataset consisted of 50 RGB field images, with corresponding colour-coded segmentation masks. Mask

colours represented background (black), crop (green), and weed (red).

Given the low amount of data available, the dataset was randomly divided using an 80/20 split, producing 40 training images and 10 validation images. This was chosen over a 70/30 split to maximise training samples and improve model learning. To organise the image-label pairs efficiently, MATLAB `imageDatastore` and `pixelLabelDatastore` were used to load these images.

Class labels were defined using RGB encoding, where background is represented by black, crop by green, and weed by red. This mapping allows the ground truth images to be converted into categorical label matrices that can be used for training.

### 2.2. Preprocessing

The original image resolution was  $1296 \times 966$  pixels. To reduce training time and memory usage, images were resized to  $240 \times 320$  during training, using MATLAB `pixelLabelImageDatastore`. This resolution preserved the overall visual structure of the images, while reducing training time and computational costs. Despite this resizing during training, the segmentation predictions were generated at the original image resolution ( $1296 \times 966$ ) during inference. This ensures the final output meets the coursework requirements.

For the improved model, data augmentation was applied to the training set. This included random horizontal reflections, small rotations, and translations. These transformations increase the diversity of the training data and help improve model generalisation.

### 2.3. Baseline CNN architecture

The baseline model was implemented as a simple fully convolutional network (FCN) designed to provide a naïve reference for comparison. The network consists of an input layer followed by two convolutional layers with  $3 \times 3$  filters and ReLU activation functions. A final  $1 \times 1$  convolutional layer maps the extracted features to the three output classes. No pooling or upsampling layers were included in the baseline architecture, allowing the spatial resolution of the input to be preserved throughout the network. The final layers consist of a softmax activation followed by a pixel classification layer, which produces per-pixel class predictions.

This architecture was intentionally kept shallow and simple, limiting its ability to learn complex hierarchical

features. As a result, it is sufficient as a baseline for evaluating the effectiveness of more advanced models.

## 2.4. Improved CNN architecture

The improved model builds on the baseline by increasing network depth and feature representation capacity. Additional convolutional layers were used, with an increasing number of filters (32 and 64), allowing the network to learn more complex and discriminative features.

Each convolutional layer uses 3×3 filters with ReLU activation, enabling non-linear feature extraction. The deeper architecture allows the model to capture richer spatial and contextual information compared to the baseline model.

Unlike more complex architectures such as encoder–decoder networks, the improved model does not include explicit downsampling or upsampling layers. This design choice simplifies the network and avoids spatial resolution mismatches, while still providing sufficient capacity to improve performance.

The final classification stage is the same as the baseline, using a 1×1 convolution followed by softmax and pixel classification layers to produce segmentation outputs.

## 2.5 Training setup

The baseline model was trained using stochastic gradient descent with momentum (SGDM), while the improved model used the Adam optimiser for faster convergence, and improved optimisation performance.

Training was performed for 15 epochs for the baseline model and 30 epochs for the improved model, with a mini-batch size of 4. The learning rate was set to 0.001 in both cases. Training data was shuffled at each epoch to improve generalisation.

Validation data was used during training to monitor performance and assess convergence. No early stopping was applied, as the number of epochs was relatively small. For the improved model, data augmentation was applied only to the training set, while validation data remained unchanged to ensure a fair evaluation.

# 3. EVALUATION

Performance was evaluated using global accuracy, class accuracy, Intersection over Union (IoU), weighted IoU, and boundary F-score (BFScore). IoU is particularly important as it measures overlap between predicted and ground truth segmentation.

## 3.1. Baseline model – results

Below are the general results of the baseline model, produced after training with 15 epochs and mini batch size of 4.

TABLE 1  
BASELINE MODEL RESULTS

| Metric | Result |
|--------|--------|
|--------|--------|

|                 |         |
|-----------------|---------|
| Global Accuracy | 0.96307 |
| Mean Accuracy   | 0.6895  |
| Mean IoU        | 0.57299 |
| Weighted IoU    | 0.93982 |
| MeanBFScore     | 0.51653 |

Table 1. General score and accuracy of baseline model

The baseline model achieves a high global accuracy of 96.3%, as shown in **Table 1**. However, this metric is misleading due to class imbalance within the dataset, where background pixels dominate. As a result, the model can achieve high overall accuracy while still performing poorly on the crop and weed classes, which is more apparent as shown in the below results.

TABLE 2  
BASELINE MODEL RESULTS – CROP AND WEED

| Metric      | Crop    | Weed    |
|-------------|---------|---------|
| Accuracy    | 0.49709 | 0.58372 |
| IoU         | 0.29917 | 0.44761 |
| MeanBFScore | 0.41453 | 0.49299 |

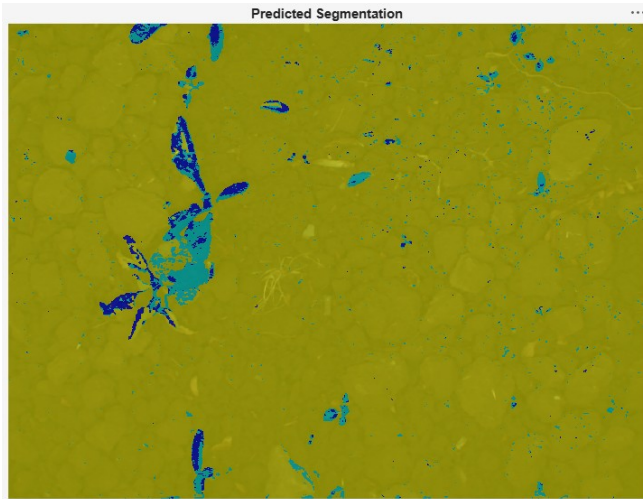
Table 2. Results of crop and weed specifically, for better understanding of class-specific performance.

**Table 2** shows the results for crop and weed categories. The model achieves moderate performance, with IoU values of 0.299 for crop and 0.448 for weed. This indicates that while the model can detect weeds reasonably well, it struggles more with accurately segmenting crop regions. The relatively low MeanBFScore values further suggest that boundary localisation is not precise.

| Confusion Matrix |            |                 |        |      |
|------------------|------------|-----------------|--------|------|
| True Class       | background | crop            | weed   |      |
|                  | 11642442   | 19845           | 125224 |      |
|                  | 20169      | 72372           | 53049  |      |
|                  | 167573     | 76477           | 342209 |      |
|                  |            | background      | crop   | weed |
|                  |            | Predicted Class |        |      |

Fig 1. Confusion matrix for the baseline model, showing significant bias towards the background class.

From the baseline model confusion matrix in **figure 1**, it can be seen that the model heavily favours the background class, with many pixels correctly classified as background, but significant misclassification occurring between crop, weed, and background, which can be seen in **figure 2**. Many crop pixels are also incorrectly predicted as background, explaining the lower IoU for the crop class in **table 2**.



**Fig 2.** Example output image of baseline model.

**Figure 2** shows an example image of the segmentation produced by the baseline model. This qualitative data demonstrates there is some reasonable classification of the crop and weed areas as it is able to capture the general regions of the plants, however, it has inconsistent boundaries. There is also overwhelming classification of background dirt, which corroborates with **Figure 1** confusion matrix. There is also some visible misclassification between the crop and weed classes.

### 3.2. Baseline model – discussion

Although the baseline model has a high global accuracy percentage, from the confusion matrix and crop and weed individual accuracies, we can see that this high percentage could be due to most of the image being classified as dirt. This is not a surprising result, as field images are typically dominated by the background soil, and given the small dataset, this high accuracy could be skewed. These limitations could also be explained by the baseline model's shallow structure, with only a small number of convolutional layers. No data augmentation was applied as well, meaning the model was trained on a very limited set of samples, leading to poor generalisation and being unable to separate the crop and weed classes clearly, as seen from **figure 2**. The baseline model also lacks more advanced architectural components, like encoder-decoder structure, which further limits its performance and accuracy.

### 3.3. Baseline model – evaluation

Despite the clear limitations mentioned in the discussions section, the baseline model achieves a result that can provide a useful reference point for an improved model, especially regarding the crop and weed accuracies. It is a decent naïve approach that can be used as a control for the improved model to compare against.

## 3.4 Improved model – results

TABLE 3  
IMPROVED MODEL RESULTS

| Metric          | Result  |
|-----------------|---------|
| Global Accuracy | 0.97039 |
| Mean Accuracy   | 0.80155 |
| Mean IoU        | 0.6477  |
| Weighted IoU    | 0.95046 |
| MeanBFScore     | 0.57096 |

**Table 3.** General score and accuracy of the improved model

The performance of the improved model is summarised in **Table 3**. Overall, improvements can be seen across all evaluation metrics, compared to the baseline model. Global accuracy saw a small increase from 0.963 to 0.970, while the mean accuracy improved significantly from 0.689 to 0.802. Most notably, the Mean IoU increased from 0.573 to 0.648, indicating a substantial improvement in segmentation quality.

TABLE 4  
IMPROVED MODEL RESULTS – CROP AND WEED

| Metric      | Crop    | Weed    |
|-------------|---------|---------|
| Accuracy    | 0.79966 | 0.61482 |
| IoU         | 0.4514  | 0.51334 |
| MeanBFScore | 0.44639 | 0.54559 |

**Table 4.** Results of crop and weed specifically of the improved model, for comparison with table 2.

Similar to the improvements shown above for **table 3**, class specific results also show an overall increase across all metrics. The most significant one being the crop accuracy, increasing from a 0.50 to a 0.80. This shows a significant gain in correctly identifying crop regions. The IoU for crop also increased from 0.299 in the baseline model to 0.451, while weed IoU improved from 0.448 to 0.513. Overall, these results indicate that the improved model is better able to distinguish the visually similar classes of crop and weed, which the baseline model struggled with. (Refer to **Figure 1** and **figure 2**.)

| Improved Model Confusion Matrix |            |        |        |
|---------------------------------|------------|--------|--------|
| True Class                      | background | crop   | weed   |
|                                 | 11671839   | 18337  | 97335  |
|                                 | 10610      | 116422 | 18558  |
| True Class                      | background | crop   | weed   |
|                                 | 131826     | 93989  | 360444 |
| Predicted Class                 |            |        |        |

**Fig 3.** Confusion matrix for the improved model, showing improved class balance and reduced misclassification compared to the baseline.

From the confusion matrix in **figure 3**, while the model still shows some of the same bias towards the background class, this bias is reduced compared to the baseline. There is a noticeable increase in correctly classifying crop pixels, as can be seen by the gain in true positives for the crop class and the reduction in misclassification as background. However, some confusion between crop and weed classes remains, despite seeing some improvements.

### 3.5 Improved model – discussion

The improved model does indicate a better ability at identifying the crop and weed classes, which could be attributed to two main factors.

First, the increased depth and number of filters in the convolutional layers allow the network to learn more complex features. This enhances its ability to differentiate between crop and weed regions, which the baseline model mainly struggled with likely due to being visually similar (in colour and texture) and having overlapping features.

Second, the addition of data augmentation played a critical role in improving generalisation. By applying random transformations such as rotations, reflections, and translations, the model is exposed to a wider variety of training examples, reducing overfitting and improving robustness, leading to the overall performance increase as seen in **table 3** and **table 4**.

However, despite these improvements, certain limitations remain. Boundary precision is still relatively low, as indicated by the MeanBFScore values, suggesting that the model struggles to accurately detect object edges. Additionally, the absence of multi-scale feature extraction or skip connections may limit the model's ability to capture fine spatial details. More advanced architectures, such as encoder-decoder networks or U-Net variants, could further improve segmentation performance. Although these architectures were not used in this project due to time constraints and limited dataset, using these networks in future implementations could greatly help performance, especially with boundary precision and separation of crop and weed classes.

### 3.5 Improved model – evaluation

Overall, the improved model successfully addressed several of the weaknesses observed in the baseline model. The combination of increased model capacity and introduction of data augmentation led to more balanced class predictions and improved segmentation accuracy, demonstrating the effectiveness of these strategies when working with limited datasets.

### 3.6 comparison and evaluation of model architecture

The differences in architecture of the baseline and improved model had a clear impact on the final results. The baseline model consists of a shallow fully convolutional architecture with only two convolutional layers and a final  $1 \times 1$  convolution for classification. This simple structure preserves spatial resolution for simplicity, but limits the model's ability to learn complex hierarchical features. As a result, it struggled to distinguish between visually similar classes such as crop and weed, as shown in **section 3.1**.

Compared to the baseline model, the improved model introduces a deeper architecture with multiple convolutional layers and an increased number of filters (32 and 64). This enables the network to learn more discriminative feature representations, improving its ability to identify variations in texture and colour. Although the model does not include explicit downsampling or upsampling layers, the increased depth alone enhanced its feature extraction ability quite significantly. Additionally, the use of data augmentation in the improved model played an important role in improving generalization, which is particularly important given the limited dataset size.

These architectural and training differences directly contributed to the observed improvements in segmentation performance. The baseline model's limited capacity leads to underfitting and strong bias towards the background class, whereas the improved model achieves better class balance and higher IoU through enhanced feature learning and improved generalisation.

## 4. CONCLUSION

This work demonstrated the effectiveness of CNN-based semantic segmentation on a small agricultural dataset. A simple baseline model was first implemented, followed by an improved model incorporating deeper architecture and data augmentation. The baseline model achieved high global accuracy but performed poorly on class specific accuracies. The improved model achieved higher scores on all metrics, which was possible due to the increased model capacity and the use of augmentation to enhance generalization. Future work could explore more advanced architectures such as U-Net or the use of pretrained models to further improve performance.

## 5. REFERENCES

- [1] MathWorks, "Semantic Segmentation Using Deep Learning," MATLAB Documentation. [Online]. Available: <https://uk.mathworks.com/help/vision/semantic-segmentation-using-deep-learning.html>
- [2] MathWorks, "pixelLabelDatastore," MATLAB Documentation. [Online]. Available: <https://uk.mathworks.com/help/vision/ref/pixellabeldatastore.html>